

A comment on “From alternative conceptions of honesty to alternative facts in communication by US politicians”

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Abstract

We replicated, as part of Münster Replication Games, a paper by Lasser et al. (2023) that analyses the shift in honesty of US politicians’ communication on Twitter between 2010 and 2022 (before the change of ownership to Elon Musk) and the implications for the quality of information shared. The original findings were that politicians increasingly emphasize “belief speaking,” which may be decoupled from evidence, distinguishing it from evidence-based “fact speaking.” For Republicans, a 10% increase in belief speaking is associated with a 12.8-point decrease in the quality of sources shared online (NewsGuard), while fact-speaking language is linked to higher-quality sources for both parties. The article included an extensive and well-structured replication package and readme file. However, we could not replicate many of the findings and figures due to incomplete data—partly because of copyright restrictions and partly due to premature deletion of the data. We could reproduce Figure 2, parts of Figure 3 and the *Welch*-test analysis of the section “Partisan and temporal dynamics of conceptions of honesty”. Based on this sections analysis we ran diverse robustness checks—including variations in sample size, statistical power, and within-person comparisons over time—confirming that these patterns were stable and not driven by specific subsamples or methodological choices. The study and the replication were observational, so causal claims cannot be made, but our findings support the hypothesis that contemporary political discourse increasingly privileges subjective belief over evidence-based fact.

1. Introduction

The article by Lasser et al. (2023) analysed the use of fact-speaking (referred to as truth-speaking in the original data set and the analyses) and belief-speaking related words in tweets of U.S. politicians over time. Belief speaking was defined as communication of “the speaker’s beliefs, thoughts and feelings, without regard to factual accuracy” (p. 2141). Fact speaking was defined as communication of “accurate information and an updating of one’s beliefs based on that information” (p. 2141). For analysis, two dictionaries were created that were descriptive of belief or fact speaking. Validity and diagnosticity were assessed by human participants. The final dictionaries consisted of 37 keywords for each component. The keywords were then converted into vector embeddings using a pre-trained algorithm (GloVe). These embeddings encode subtle contextual information and allow a quantification of the similarity between each dictionary and a given text through vector-based comparison.

For each tweet a belief-speaking similarity score (D_b) and a fact-speaking similarity score (D_f) was calculated. A positive similarity score indicates that the text is more closely related to the terms contained in the given dictionary; a negative similarity score shows the opposite. The authors quantified the change in belief speaking and fact speaking over time by comparing the mean similarity scores of politicians’ tweets across two periods. An average belief-speaking score and an average fact-speaking score were calculated for each twitter account. These account-level averages were then grouped by party and by time period (2010-2013 and 2019-2022; note that the reported time windows differ in the article and the provided code). Lasser et al. conducted unpaired *Welch*-tests separately for Democrats and Republicans to assess whether these means differ between the early and the late period. The *Welch*-tests showed that both belief-speaking and fact-speaking scores were significantly higher in the later period for both the Republicans and the Democrats.

However, both parties differed considerably regarding the quality of shared information. Republicans shared information of lower quality compared to Democrats. This effect is associated with belief speaking: The more Republicans used belief-speaking language, the more likely they were to circulate information of low-quality. For Democrats, no reliable evidence for such a relationship was found.

We decided to replicate all claims made in the original article. To do so, we first examined the authors’ documentation of their analyses. We then verified whether all data and the corresponding code were fully available. During this process, we discovered that the raw dataset could not be provided due to external restrictions beyond the authors’ control. Our own attempts to independently obtain these data were unsuccessful, which limited our ability to reproduce the full set of extensive analyses. As a result, we were only able to replicate a subset of the original findings.

2. Computational Reproducibility

We used the extensive replication package provided by the authors on Github (<https://github.com/JanaLasser/new-ontology-of-truth/tree/main>) and the data from the OSF repository (<https://osf.io/vny8k/overview>). The paper comprises an impressive number of in-depth analyses, many of which are reported in the supplementary material. Because of the time-constraints of one day, we focused on replicating the results of the analyses in the main manuscript and a select number of supplementary analyses.

The README file states that the original tweet texts from Twitter could not be provided in the repository for data protection reasons. The authors provide code to acquire this data via the Twitter/X API but, depending on the level of X.com developer account, it is only possible to download a limited number of tweets per hour or day. For us, with a free account, there would have been a limit of 100 posts a day. Because the full data comprise around 2.7 m posts, this was not feasible.

We tried to run the code that outputs the file required for the subsequent analyses (“hydrate_tweets.ipynb”), on the 100 posts that we could download with a free X account. This failed because of code errors, possibly because of version discrepancy in some of the Python packages used in the analysis (Lasser et al. state the R and Python versions used in their analysis, but not those of single modules or packages).

The README file states also that most of the analyses can still be run because the repository provides data files that omit the tweet texts and other data that the authors are not allowed to share, but should include the values needed to calculate all necessary statistics. An intermediary file that omits the tweet text but has all the data to produce the files needed for analysis seems to have been removed from the OSF repository (“combined_US_politician_twitter_timelines_2010-11-06_to_2022-12-31_clean.csv.gz”). The two files “tweets_for_lme_modelling_NG.csv.gz” and “tweets_for_lme_modelling_independent.csv.gz” were missing as well. They appear to have been deleted from the OSF repository. These files are necessary for reproducing the linear mixed effects models that test the association between belief-speaking and fact-speaking similarity, party affiliation, and trustworthiness of linked articles.

We aimed to reproduce the figures from the article. As stated in the README file, the replication of Figure 1 was not possible because the original twitter posts were required. Figure 2 could be exactly replicated by using the original script.

Contrary to what is stated in the readme file, the reproduction of the linear mixed effects model was not possible, because the code referred to the data “combined_US_politician_twitter_timelines_2010-11-06_to_2022-12-31_lemma.csv.gz” derived from the raw data set. The computed data set was removed from the repository Mar 11, 2023.

The reproduction of the prediction lines in Figure 3 was only partly possible because data from this linear mixed-effects model were required. We could only reproduce Figure 3 to this extent (Figure 1).

Figure 1

Partial Reproduction of Figure 3 of p. 2144

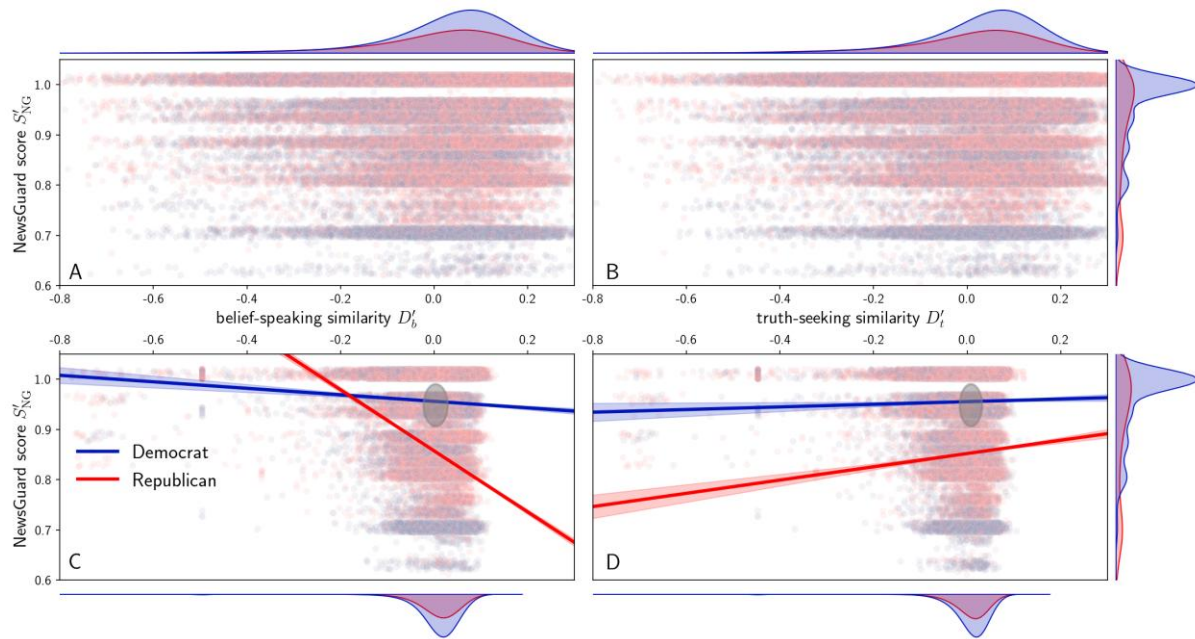


Table 1 lists the files from the tweet_collection folder, their function, and reproducibility. The files from this folder are used to scrape and process the twitter handles and tweets and to prepare them for later analysis. Because we were not able to download the tweets, we were not able to test the reproducibility in this case.

Table 1

Files and reproducibility from tweet_collection folder

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
get_US_politician_twitter_account.ipynb	Scrape Twitter handles for politicians from different web sources	none	script does not complete	HTTP error for one of the websites from which handles are scraped (URL parameters changed?)
get_US_politician_twitter_timelines.ipynb	Scrape tweets from politicians from Twitter	none	script does not run	author's server hard-coded into script

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
hydrate_tweets.ipynb	recreate tweet texts from tweet IDs	tweet_IDs.txt	script does not complete	depends on Twitter API, code error in “extract_urls” function
wrangle_data.ipynb	create files for statistical analysis	multiple files created by “hydrate_tweets.ipynb”	could not run	indicated in the README: “requires NewsGuard data base and tweet texts”

Table 2 lists the files from the article_collection folder and their reproducibility. In the readme file, it is indicated that corpus of articles from the New York Times cannot be published. We thus checked the scripts, but could not directly test them on the articles.

Table 2

Files and reproducibility from article_collection folder

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
article_pre-processing.r	Processes linked articles to provide structured file for statistical analysis	data/articles/article_corpus_raw.rds	No	File not provided. Indicated in the readme that New York Times articles cannot be shared.
article_scraping.ipynb	Unknown, not described in README		Not tested	Some paths referring to author’s computer hardcoded in script.
article_scraping.r	Produces data frame with articles linked in tweets and outputs it into a csv file	data/articles/url_list_for_article_scraping.csv	No	File provided in repository, but “score” column of NewsGuard scores does not exist
article_script.py	Helper script for article_scraping.r	Fetches article texts from URLs in tweets	Not tested	called by article_scraping.r

Table 3 lists the files from the analysis folder and their reproducibility. The scripts and analyses from this folder could in part be reproduced. Others relied on the original tweet texts or news articles, which authors could not share, and some referred to datafiles which had been removed from the repository.

Table 3*Files and reproducibility from analysis folder*

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
bertopic_model.ipynb	modelling of controversial topics (Supplement 6)		not attempted (part of Supplement 6)	indicated in the README: “requires tweet texts”
bootstrapping.ipynb	Bootstrapping for confidence intervals		not attempted, output files are in repository	
compute_sbert_avg_lexicon.py	calculating honesty component similarity for the different corpuses (NYT articles, tweets and linked articles)	tweets/combined_US_politician_twitter_timelines_2010-11-06_to_2022-12-31_clean.csv.gz	No, file not provided	
compute_sbert_avg_lexicon_reduce_lexicon_loop.py	calculating keyword similarity, called by label_lexicon_loop.sh		Not attempted, helper script	
compute_sbert_avg_lexicon_reduce_lexion_single_word.py	assessing dictionary robustness, called by label_lexicon_single_word.sh		Not attempted, helper script	

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
compute_VADER_scores.ipynb	assess sentiment of posts using the VADER model	combined_US_politician_twitter_timelines_2010-11-06_to_2022-12-31_lemma.csv.zip	No, file not provided	
create_document_level_validation_sample.ipynb	Create sample for document_level_validation.R	US_politician_tweets_2010-11-06_to_2022-03-16.csv.zip	No, file not provided	
descriptive_dataset_statistics.ipynb			partial (t-Test for heading 2)	indicated in the README: “requires NewsGuard data base”
dictionary_robustness.ipynb	Dictionary robustness validation analyses	US_politician_tweets_2010-11-06_to_2022-03-16.csv.zip	No, file not provided	
document_level_validation.R	Figures 3,4,5, &6 in the Supplementary material	data/validation/document_validation_data.csv data/validation/document_validation_sample.csv	Yes there were differences some boxplots: Supp. Fig. 4 ID 1 & 2, Supp. Fig. 5 ID 3 and change of title for both figures	changes: line 16, (“responseld, 22:143” to “responseld, 22:139”)
keyword_validation.R	validation of belief and fact keywords from questionnaire data (Validation step 1)	data/validation/validation_belief.csv data/validation/validation_truth.csv data/utilities/belief_speaking_p=0.05_swapped_wn_def_example.csv	Yes Replication of Supp. Tab. 1 and Supp. Fig. 1 & 2	changes: create long format data frame for the t-tests

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
	Additionally provides density plots not reported in the article	data/utilities/truth_seeking_p=0.05_sw apped_wn_def_example.csv		
label_fasttext- cc_DDR.sh	Calculate honesty component similarity	Tweet texts	Not attempted	indicated in the README: “requires tweet texts”
label_glove840B_DDR. sh	Calculate honesty component similarity	Tweet texts	Not attempted	indicated in the README: “requires tweet texts and New York Times abstracts”
label_lexicon_loop.sh	Reproduction of dictionary robustness analysis	tweets/combined_US_politician_twitter _timelines_2010-11-06_to_2022-12- 31_clean.csv.zip and tweet texts	Not attempted	indicated in the README: “requires tweet texts”
label_lexicon_single_ word.sh	Keyword similarity score	Tweet texts	Not attempted	indicated in the README: “requires tweet texts”
label_word2vec- googlenews_DDR.sh				indicated in the README: “requires tweet texts”
Imer_models_tweets. Rmd	Linear mixed effects models for analysis of tweets	tweets/tweets_for_lme_modelling_NG. csv.zip tweets/tweets_for_lme_modelling_inde pendent.csv.zip	No, files not provided	

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
mediation.Rmd		data/users/users.csv	yes	it was an analysis of the supplement therefore we did not work with it
OLS_regression_articles.ipynb	Produce estimates for belief speaking and fact speaking for articles linked in tweets.	data/articles/articles.csv.zip	yes	Data file has to be unzipped and “tables” folder created first
plots.ipynb	Produces Figures 2 and 3.	users/users.csv (provided) bootstrapping/belief.csv (provided) bootstrapping/truth.csv (provided) tweets/tweets.csv.zip (provided) tweets/LME_predictions_tweets_belief.csv (not provided, although the README states it should be) lmer_models_tweets.Rmd data/articles/OLS_predictions_articles (generated by OLS_regression_articles.ipynb)	Partially. Figure 2 is created appropriately, Figure 3 is only created partially (without prediction lines from linear mixed effects models).	depends on output from lmer_models_tweets.Rmd
scattertext.R	Figure 1	data/tweets/combined_US_politician_twitter_timelines_2010-11-06_to_2022-12-31_lemma.csv.zip	No, dataset combined_US_politician_twitter_timelines_2010-11-06_to_2022-12-	indicated in the README: “requires tweet texts”

Code File	What does it do?	Relies on which data file	Reproducibility	Explanation
		/data/utilities/marimo.txt	31_lemma.csv.gzipnot provided	
		data/utilities/stopwords_en.txt		
		data/utilities/belief_speaking_p=0.05_ swapped_wn_def_example.csv		
		data/utilities/truth_seeking_p=0.05_ swapped_wn_def_example.csv		
scrub_data_for_ upload.ipynb	Scrubbing protected information from data sets for upload to OSF	US_politician_tweets_2010-11- 06_to_2022-03-16.csv.gzip	No, file not provided	
validation.R	same as keyword_validation.R	data/validation/validation_belief.csv data/validation/validation_truth.csv data/utilities/belief_speaking_p=0.05_ wapped_wn_def_example.csv data/utilities/truth_seeking_p=0.05_sw apped_wn_def_example.csv	Yes	

We focused our reproduction primarily on the section “Partisan and Temporal Dynamics of Conceptions of Honesty,” as the data and code were provided there (Table 4). Reported for T1 was 2011-2013, however calculated was 2010-2013. 2010-2013 is probably correct as it should be 4 years. Our results fit the reported numbers in the paper. In the code and data repositories the honesty component that is called fact speaking in the paper is instead called truth speaking.

Table 4

Analysis of the Welch Test p. 2142

Step	Reproducibility	Code file	Relies on which data file	Explanation
from raw data	no	no	no	
from analysis data	yes	descriptive_dataset_statistics.ipynb	data/users/users.csv	

2.1 Discrepancies Between Pre-analysis Plan and Article

To our knowledge, the authors did not register a pre-analysis plan.

3. Robustness Reproduction of the Section “Partisan and Temporal Dynamics of Conceptions of Honesty”

We implemented a robustness reproduction with the statistical program R designed to evaluate whether the core findings of the original *Welch* test analysis on p. 2142 remain stable under alternative specifications, sample definitions, and statistical approaches. Each modification was introduced separately to clarify its specific impact on the main point estimates; the checks were motivated by potential sensitivities in sample size, statistical power, and dependency structures in the data.

3.1. Change of Sample Size

First, we reduced the sample for all four *Welch* tests to a smaller subset of 150 politicians (seed = 123, Table 5 Middle) to assess whether the original results (Table 5 Top) depend on the full dataset or whether they can be reproduced with sample sizes comparable to smaller countries. The effect estimates remained statistically significant, indicating that the findings are not driven by sample size alone.

Table 5*Welch Test Results of Different Sample Sizes*

Data base	Group	<i>t</i>	<i>df</i>	<i>p</i>	<i>d</i>	CI of <i>d</i>	
						upper	lower
Original Sample Sizes: 2010–2013: 331 Democrats and 514 Republicans and 2019–2022: 295 Democrats and 494 Republicans							
	Democrats belief	-11.317	558	< .001	-0.850	-0.994	-0.706
	Republicans belief	-10.819	493	< .001	-0.854	-1.004	-0.703
	Democrats fact	-9.753	516	< .001	-0.748	-0.891	-0.605
	Republicans fact	-8.442	483	< .001	-0.671	-0.819	-0.522
Sample Size: 150							
	Democrats belief	-7.126	105	< .001	-0.772	-1.031	-0.513
	Republicans belief	-6.668	153	< .001	-0.958	-1.242	-0.674
	Democrats fact	-5.566	179	< .001	-0.744	-1.003	-0.486
	Republicans fact	-5.244	140	< .001	-0.773	-1.052	-0.494
Sample Size: 100							
	Democrats belief	-5.673	67	< .001	-0.773	-1.052	-0.494
	Republicans belief	-5.056	74	< .001	-0.989	-1.348	-0.630
	Democrats fact	-4.555	100	< .001	-0.989	-1.348	-0.629
	Republicans fact	-4.077	76	< .001	-0.793	-1.146	-0.441

To further verify adequacy of sample power, we calculated optimal sample sizes required to achieve a test power of 95% at the lower confidence interval of Cohen's *d* of the initial *Welch* test. The resulting sample sizes were 99, 53, 72, and 96. These numbers imply that even substantially smaller

subsets would be sufficient to detect the estimated effects. Consistent with this, restricting the sample to exactly 100 observations still produced statistically significant results (seed = 123; Table 5 Bottom).

These checks collectively show that the main conclusions are not sensitive to moderate reductions in sample size or achievable statistical power. Practically, this means that the observed effects are sufficiently strong to be detectable in smaller democracies and are unlikely to be mere artifacts of very large datasets that are typically available only in large countries such as the United States.

3.2 Adding an Individual Perspective to the Analysis

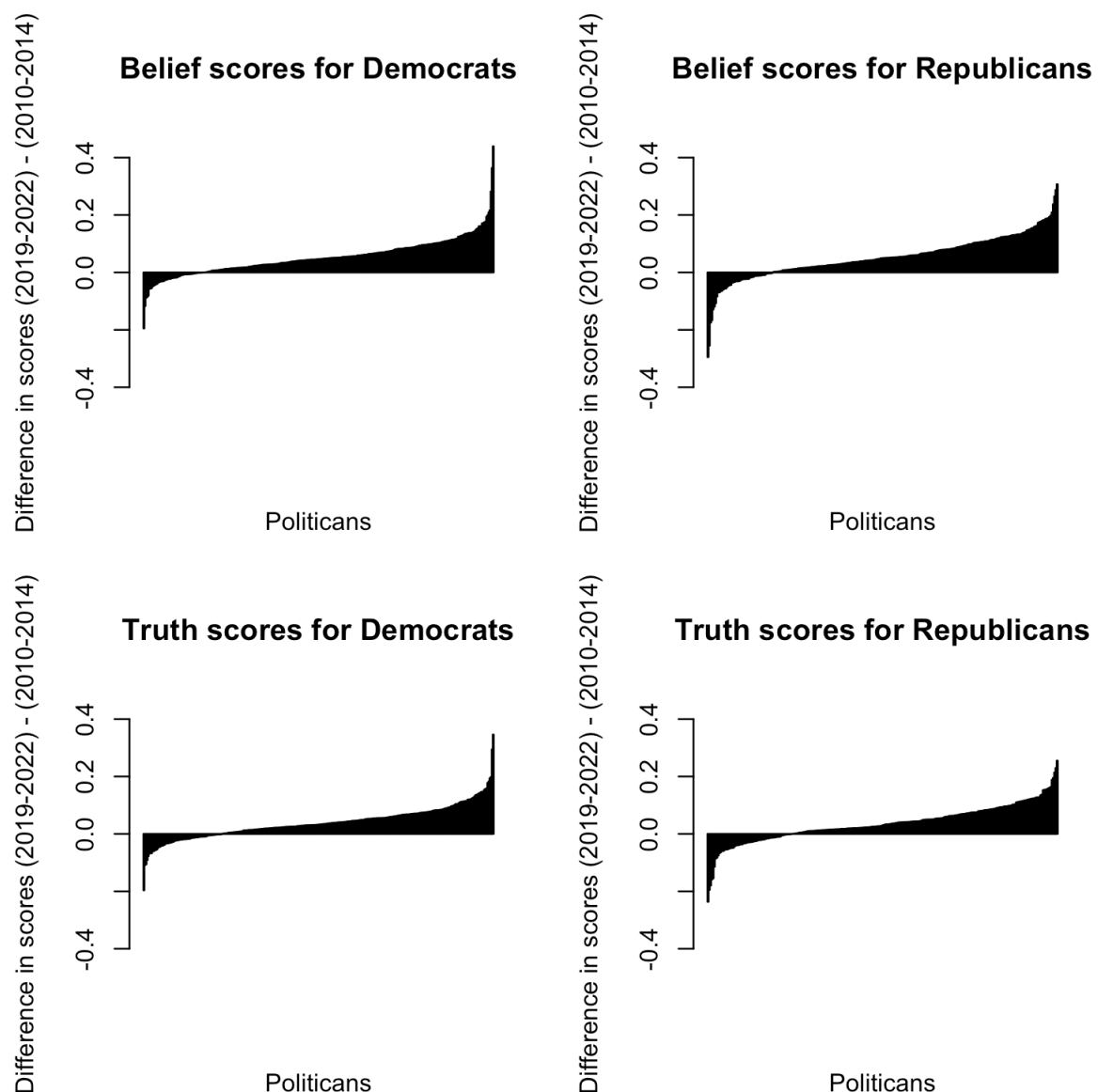
Given that many politicians (320 Democrats, 276 Republicans) appear in the data twice, we also estimated paired-sample *t*-tests after restricting the dataset to politicians observed in both periods. All tests remained statistically significant (Table 6).

Table 6
*Results of a Paired *t* Test*

Group	<i>t</i>	<i>df</i>	<i>p</i>	<i>d</i>	CI of <i>d</i>	
					upper	lower
Democrats belief	-14.549	31	< .001	-0.959	-1.115	-0.803
Republicans belief	-11.530	27	< .001	-0.786	-0.940	-0.633
Democrats fact	-11.943	31	< .001	-0.813	-0.967	-0.659
Republicans fact	-9.017	27	< .001	-0.608	-0.752	-0.464

This strengthens the argument that the original effects are not driven by compositional differences between cross-sectional samples but instead hold when examining within-person change. The implication is that the effect captures temporal variation rather than population turnover.

The sorted individual differences in Figure 2 show consistent directional shifts across both partisan subsamples and both scores and provide an additional lens on the data that goes beyond partisan affiliation.

Figure 2*Individual Differences of Politicians*

3.3 Changing the Time Period

Due to the lack of raw data, we were unable to assess the robustness of the four-year interval. However, since four years correspond to the typical legislative term in most countries, this choice appears reasonable.

3.4 Changing the Analysis Model

We also fitted a multilevel model using the initial dataset to account for the individual as a source of variation. Only Democrats and Republicans were included in the analysis (the original data

includes some independent and Libertarian party politicians). The intra-class correlation was 12.016 % for belief-speaking similarity and 17.361 % for fact-speaking similarity of politicians' tweets, indicating that a substantial proportion of the variance is attributable to differences between individuals, which justifies the use of a multilevel model to appropriately model these dependencies. Therefore, we fitted a linear mixed-effects model predicting belief speaking as well as fact speaking similarity scores from party, time, and their interaction, including a random intercept for individuals to account for repeated measures (Tables 7, 8).

Table 7

Results of Mixed Effects Models with Belief-speaking Similarity as DV Including Person as a Random Effect

<i>Predictors</i>	Model for belief speaking without number of tweets			Model for belief speaking with number of tweets		
	<i>Estimates</i>	<i>CI</i>	<i>p</i>	<i>Estimates</i>	<i>CI</i>	<i>p</i>
(Intercept)	-0.033	-0.039 – -0.026	< . .001	-0.036	-0.043 – -0.028	< . .001
party [Republican]	-0.008	-0.017 – 0.001	0.100	-0.007	-0.016 – 0.002	.146
time [2019–2022]	0.049	0.042 – 0.056	< . .001	0.049	0.043 – 0.056	< . .001
party [Republican] × time [2019–2022]	0.003	-0.007 – 0.013	.599	0.003	-0.007 – 0.013	.606
N tweets				0.001	-0.000 – 0.001	.109
ICC	0.320			0.320		
N	1038 _{id}			1038 _{id}		
Observations	1634			1634		
Marginal R ² / Conditional R ²	0.151 / 0.423			0.152 / 0.423		

Table 8

Results of Mixed Effects Models with Fact-speaking Similarity as DV Including Person as a Random Effect

<i>Predictors</i>	Model for fact speaking without number of tweets			Model for fact speaking with number of tweets		
	<i>Estimates</i>	<i>CI</i>	<i>p</i>	<i>Estimates</i>	<i>CI</i>	<i>p</i>
(Intercept)	-0.028	-0.033 – -0.023	< . .001	-0.034	-0.041 – -0.028	< . .001
party [Republican]	-0.010	-0.017 – -0.002	. .016	-0.008	-0.016 – -0.000	. .048
time [2019–2022]	0.037	0.031 – 0.043	< . .001	0.038	0.032 – 0.044	< . .001
party [Republican] × time [2019–2022]	-0.002	-0.011 – 0.007	.666	-0.002	-0.011 – 0.007	.650
N tweets				0.001	0.001 – 0.002	< . .001
ICC	0.303			0.303		
N	1038 _{id}			1038 _{id}		
Observations	1634			1634		
Marginal R ² / Conditional R ²	0.116 / 0.383			0.124 / 0.384		

For both outcomes, the results show a significant overall increase in both belief-speaking and fact-speaking similarity over time (2019–2022) compared to the earlier period (2010–2013), with an estimated increase of 0.05 points for belief speaking and 0.04 points for fact speaking, thus replicating the results of the *Welch* tests.

For belief speaking, there was no significant difference between Democrats and Republicans in the temporal trend, nor a significant interaction effect, indicating that both parties exhibited similar increases over time. Nearly a third of the variance (32%) was attributable to individual differences.

For fact speaking, there was a significant difference between the two parties, with Republicans exhibiting around 0.01 points less fact speaking overall, but no interaction between party and time, again suggesting that increases are comparable for both parties over time. Again, around a third of the variance (30 %) was attributable to individual differences.

As a robustness check, we entered the number of tweets (scaled to 1 = 1,000 tweets) that a politician had posted on average during these timespans as a covariate, which did not change the significance or direction of these estimates. For fact speaking only, the covariate achieved significance. Entering the number of tweets reduced the estimate for party affiliation, suggesting a smaller decrease

for fact speaking for Republican politicians (0.08 points compared to 0.1 points in the model without number of tweets). The results still broadly confirm the original *Welch* test analysis.

4. Conclusion

Table 9

Provided data and code

	Fully	Partial	No
Raw data provided			x
Cleaning code provided	x		
Analysis data provided		x	
Analysis code provided	x		
Reproducibility from raw data			x
Reproducibility from analysis data		x	

All robustness checks, without exception, confirm the stability of the original estimates. Together, they suggest that the primary conclusions – that belief speaking and fact speaking simultaneously increase – withstand variations in sampling, statistical power, and model specification, reinforcing confidence in the reliability of the replicated findings.

The original authors have included an extensive replication package and readme file and seem to have invested extensive thought and work into making the analyses replicable. We suggest adding codebooks that explain the meaning of all variables (e.g., for the file `users.csv`) and a guide indicating which data and code files correspond to which parts of the article. We understand that including all raw data in the replication package was not possible due to legal restrictions (Table 9). Additionally, access to data from Twitter/X has become much more restricted, with stricter rate limits for free accounts, and academic accounts seem to have been discontinued. However, we suggest providing coded or anonymized data that do not contain the actual content, and adjusting the code accordingly, so that readers can reproduce the analyses.

5. Response of the Original Authors

The authors responded in-depth to this report. They welcomed the feedback and pointed out the inherent difficulty of reproducing complex and computationally intensive analyses as the ones done for their article. They also provided a list of updates that they have made to their original replication packages in response to our report. They provided additional files in the repository that should enable the replication of more of the analyses and figures in the original article. As some of the

data used in the original article can only be accessed with a subscription, these cannot be made part of the replication package. We confirmed the updates of the file but did not have the opportunity to re-run the analyses after the revisions to the replication package.

References

Lasser, J., Aroyehun, S. T., Carrella, F., Simchon, A., Garcia, D., & Lewandowsky, S. (2023). From alternative conceptions of honesty to alternative facts in communications by US politicians. *Nature Human Behaviour*, 7(12), 2140–2151. <https://doi.org/10.1038/s41562-023-01691-w>